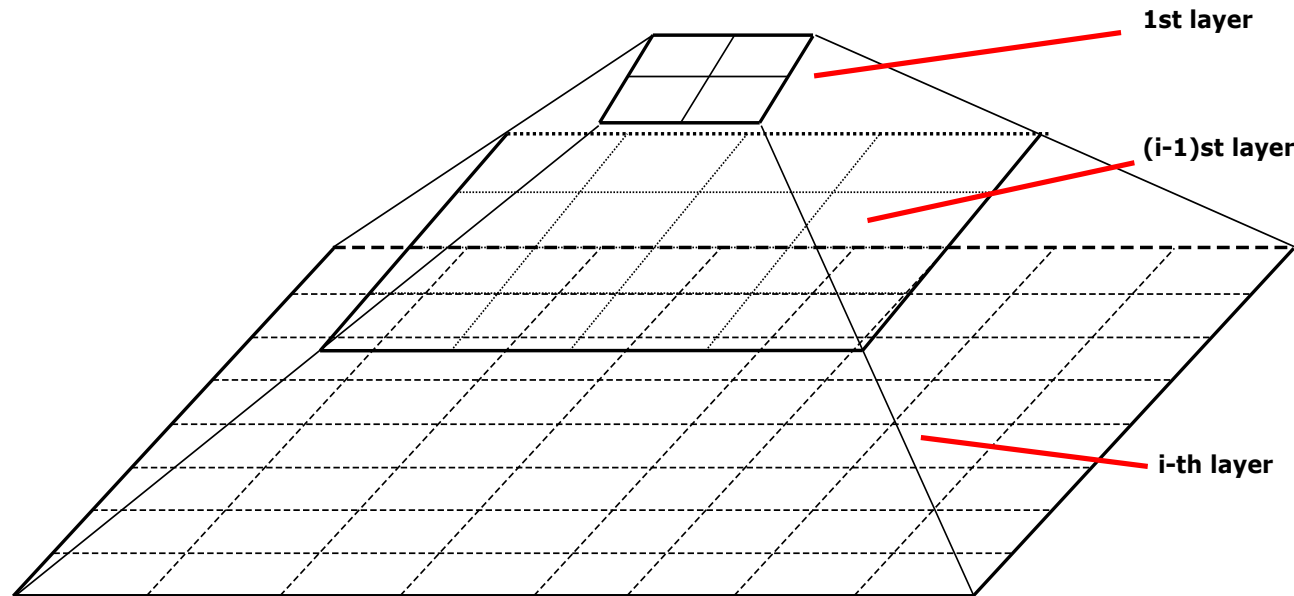


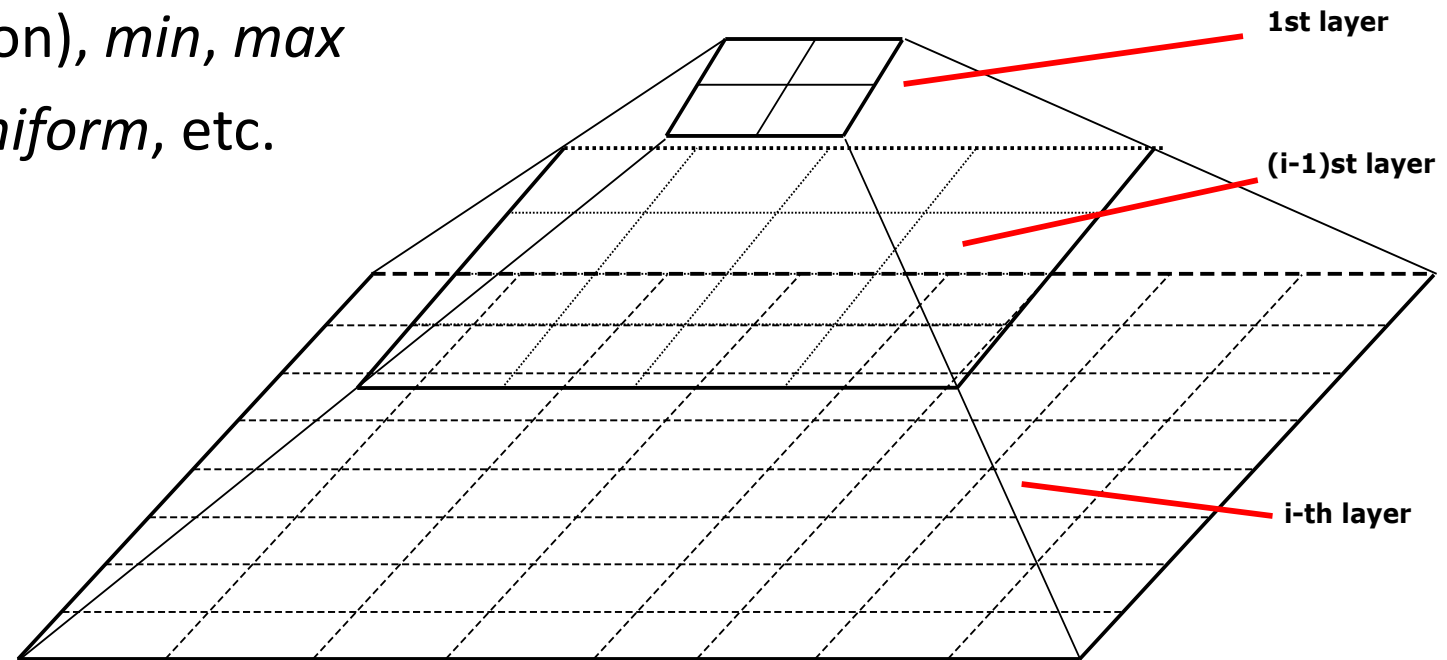
STING: A Statistical Information Grid Approach

- ❑ STING (Statistical Information Grid) (Wang, Yang and Muntz, VLDB'97)
- ❑ The spatial area is divided into rectangular cells at different levels of resolution, and these cells form a tree structure
- ❑ A cell at a high level contains a number of smaller cells of the next lower level



STING: A Statistical Information Grid Approach

- ❑ Statistical information of each cell is calculated and stored beforehand and is used to answer queries
- ❑ Parameters of higher-level cells can be easily calculated from that of lower-level cell, including
 - ❑ *count, mean, s*(standard deviation), *min, max*
 - ❑ type of distribution—*normal, uniform*, etc.



Query Processing in STING and Its Analysis

❑ To process a region query

- ❑ Start at the root and proceed to the next lower level, using the STING index
- ❑ Calculate the likelihood that a cell is relevant to the query at some confidence level using the statistical information of the cell
- ❑ Only children of likely relevant cells are recursively explored
- ❑ Repeat this process until the bottom layer is reached

❑ Advantages

- ❑ Query-independent, easy to parallelize, incremental update
- ❑ Efficiency: Complexity is $O(K)$
 - ❑ K : # of grid cells at the lowest level, and $K \ll N$ (i.e., # of data points)

❑ Disadvantages

- ❑ Its probabilistic nature may imply a loss of accuracy in query processing